Video Analysis of a Fruit-Fly Pair

Position, sex, facing direction, and male wing angles

Task and dataset

- One male and one female in a circular well.
- 326 annotated frames from one recording, with body points, heads, abdomens, and male wing tips.
- Five video excerpts at 1530×1530 and 35 fps.
- Random frame split: 218 training, 108 test. Seed 0.
- Outputs: annotated video, detection CSV, and wing-angle plot.

Python implementation based on Steven Herbst's CS229 pipeline (2018).

Four-stage method

1. **Body detection.** Threshold the frame. A decision tree classifies contour area as zero, one, or two flies. Keep one-fly contours and use their centroids as positions.
2. **Sex.** Logistic regression compares both contours' areas and aspect ratios. It runs only when exactly two separate flies remain.
3. **Heading.** Image moments give a body axis. HOG features, 15 PCA components, and logistic regression resolve the 180° ambiguity.
4. **Wings.** Rotate the male upright. HOG features and 40 PCA components feed one linear regressor for both wing halves. Clip angles to 0–90°.

Individual model results

Stage	Test metric	Result	Examples
Body count	Classification error	0%	329 contours
Sex	Classification error	0%	97 pairs
Heading reversal	Classification error	0%	152 flies
Wing angle	Residual SD	7.26°	158 wings

Annotated counts select the eligible examples. The wing model uses annotated headings for alignment. These tests do not include all errors from preceding stages. Residual SD differs from mean absolute error.

Source: `outputs/train_metrics.json` and `src/train.py`.

Complete pipeline results

194 of 216 labeled flies matched across 108 test frames.

- Sex accuracy: 100% on 194 matches.
- Mean heading error: 1.60° on 152 labeled headings.
- Mean absolute wing error: 5.87° on 158 wings.
- Mean distance from centroid to body label: 33.91 pixels.

The evaluator accepts detections within 80 pixels of a body label. It excludes 22 unmatched flies from these accuracy and error values. Body labels mark interior points rather than silhouette centroids.

Source: `outputs/train_metrics.json`.

Runtime and limits

- Saved demo throughput: 66.05 fps on `test4.mp4`, compared with a source rate of 35 fps.
- Timing includes video I/O, annotation, and CSV writing. It excludes the final plot and does not measure live-camera latency.
- Merged bodies are skipped. The program has no tracking state between frames.
- Training and test frames share one recording. Performance on new pairs or recording conditions is untested.
- Wing-angle predictions have no independently validated song labels.

Sources: `outputs/demo_metrics.json`, `src/demo.py`, and `src/pipeline.py`.